

How did the Maya civilization react to climate change?

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Abstract

Do you choose to do different activities on rainy days and sunny days? Much like we react to the weather day to day, humans also respond to the climate. The climate is the long-term pattern in weather. Changes in climate, such as long droughts, can affect how people live. It can even impact whole civilizations.

We wanted to find out whether climate change affected the Classic Maya civilization. This civilization lived in

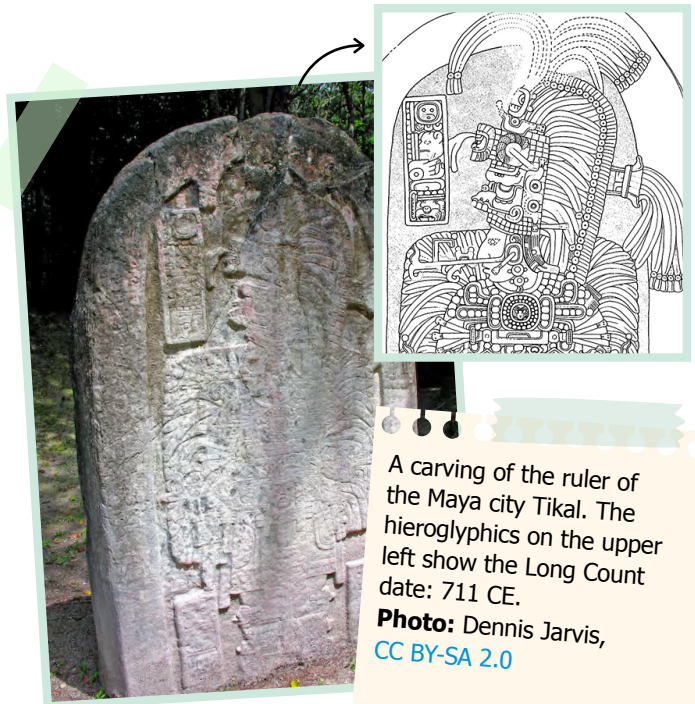
Mesoamerica over a thousand years ago. We studied the layers of a stalagmite pillar from a cave. These layers showed evidence of long, severe droughts. Soon after, the Maya people abandoned their cities in the area. We think the droughts might have forced people to leave and caused their political systems to collapse.

Introduction

Can you imagine almost no rain falling for 13 years? Even with modern technology, life is difficult without rain. Farmers can't grow crops, and there isn't enough drinking water.

People living in the ancient **Maya civilization** may have faced severe **droughts**. The Maya civilization peaked in **Mesoamerica** more than a thousand years ago. They built towering temples, used advanced math, and wrote in hieroglyphics. They lived in city-states, each with its own warrior-king and a council of advisors. They had a calendar system called the **Long Count** and carved important dates on monuments and buildings.

Maya civilization changed a lot between the years 800 and 1000 **CE**. We call this period the **Maya Terminal Classic**. Historians and **archaeologists** noticed that there were fewer Long Count dates on monuments during this time. They also found that people stopped building and making things. This was based on **radiocarbon dating**. We think that the Maya people abandoned their cities to move further north, where they built new places to live.



A carving of the ruler of the Maya city Tikal. The hieroglyphics on the upper left show the Long Count date: 711 CE.

Photo: Dennis Jarvis,
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Why did this happen? Changes in the **climate**, such as long droughts, may have caused crops to fail and people to go hungry. But it's hard to know if climate change caused these cities to collapse. We don't have a written record of the climate from the period.

Luckily, we have a natural record! In caves, there are mineral columns called **stalagmites**. When it rains, water seeps through the cave roof and drips into the cave. This water is called **drip water**. Stalagmites grow taller as

drip water leaves behind layers of **calcite**. This is like a tree growing a new ring every year. Each new layer in the stalagmites holds information about the climate in the year it formed.

We wondered if we could see evidence of droughts in **stalagmites**. We wanted to compare this record with the human history of the Classic Maya civilization. This could help tell us more about the effects of drought on the Maya people.

Methods

The stalagmite we studied (Figure 1A) came from a cave in the Yucatán Peninsula in Mexico. This would have been near two major ancient Mayan cities.

First, we had to work out the age and date of the stalagmite.

- We ground up small sections into a powder. We measured chemical **isotopes** of the elements uranium and thorium. These isotopes change over time. By studying how much they have changed, we can figure out the age of the rock or object.
- We cut the stalagmite in half and used a video microscope to count all the calcite layers inside.

Next, we looked for evidence of drought in the stalagmite layers.

- We used chemical analysis to measure oxygen isotopes in the layers. This tells us how rainy or dry it was when each calcite layer formed.
- Every month for a year, we measured the amount of rainwater and collected drip water (Figure 1B). We analyzed the oxygen isotopes in the drip water. This helped us check that changes in oxygen isotopes matched changes in the amount of rainfall.

Finally, we compared our climate data to Long Count dates on carved monuments in the region.

A.



B.



Figure 1:

A. A stalagmite forming in the cave where we found our stalagmite. The stalagmite we studied broke off naturally, and we collected it from the cave floor. This is the best way to preserve the cave environment.

B. Ola Kwiecien and our drip water collection system. The box collects the water that seeps through the cave roof and drips into the cave where the stalagmite formed.

Photos: Sebastian Breitenbach

Results

We found that part of the stalagmite spans the years 871-1021.

- This means it was forming during the period when Mayan civilization was changing.
- There were around 180 layers in the stalagmite. Each layer represents a year.

We found evidence of drought in the stalagmite layers.

- Our oxygen isotope measurements of the calcite layers showed eight extreme droughts of three or more years

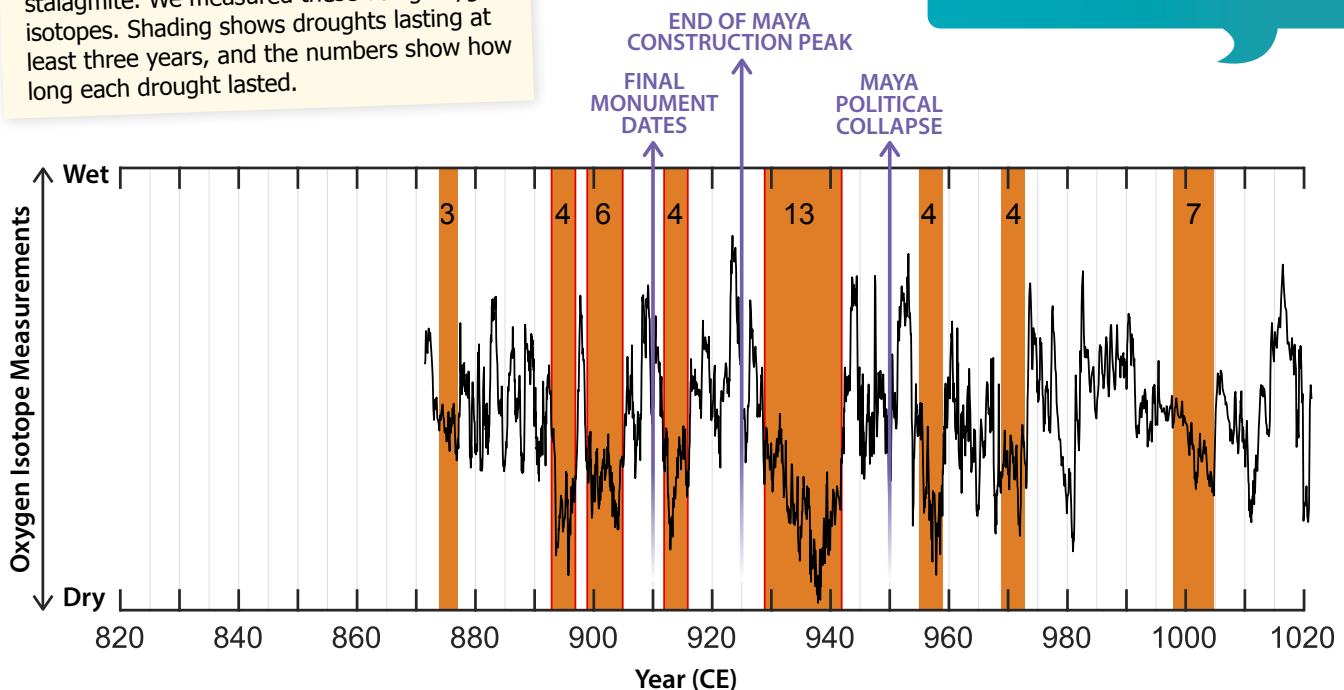
in a row. This included a 4-year drought starting in 894 CE and a 13-year drought from 929 to 942 CE (Figure 2).

- The drip water we collected showed that our oxygen isotope measurements matched the rainy and dry seasons.

We know from Long Count dates that the Maya people did not build many monuments after 900 CE. They had stopped building in this region by 925 CE. This lines up with the droughts in the late 800s and early 900s.

Figure 2: We recorded the dry years in the stalagmite. We measured these using oxygen isotopes. Shading shows droughts lasting at least three years, and the numbers show how long each drought lasted.

What was the longest drought recorded in the stalagmite, and when did it take place?



Discussion

We learned that there were many long droughts in the region in the late 800s and early 900s. Not long after this, the Maya stopped building temples and monuments. Historians and archaeologists know from studying Long Count dates that the Classic Maya political system had collapsed by 950. This was 8 years after the end of the 13-year drought we found.

The Maya had many strategies to deal with drought, like water wells and canals. But we think repeated long droughts were serious enough to cause crop failure and hunger. The droughts may also have led to conflict and disease, bringing political instability.

Conclusion

Studying the stalagmite gave us a natural record of droughts that happened in the 800s and 900s. Major political changes in the Maya civilization happened around the same time. This shows that our scientific evidence supports what historians already believed.

In school, science and history are often taught as separate subjects. But when we use them together, they can help us learn even more. Scientists can study rocks, tree rings, ice, or even old bones to understand what life was like long ago. What part of history interests you most? How could you use science to learn more about it?

Glossary of Key Terms

Archaeologists - scientists who study human history by digging up and examining objects, buildings, and other remains from the past.

Calcite - a hard mineral made from calcium, carbon, and oxygen. It forms when water leaves behind minerals in caves, slowly building features like stalagmites and stalactites.

Climate - the usual pattern of weather in a place over a long time. Climate includes things like average temperature, rainfall, and seasonal changes.

Common Era (CE) - the dating system widely used today to count years. "CE" means Common Era and uses the same numbering as AD.

Drip water - water that slowly drips from the ceiling of a cave, often carrying minerals that can form stalactites and stalagmites.

Drought - a long period when a place gets much less rain than normal. Droughts can dry up water sources and make it difficult for plants, animals, and people to get enough water.

Isotopes - slightly different forms of the same chemical element. Scientists can measure isotopes to learn details such as the age of materials or where water or minerals came from.

Long Count - a system of counting years used by the ancient Maya. It helped them track long periods of time and record important events.

Maya civilization - an advanced society that lived in parts of Mexico and Central America for thousands of years. The Maya are known for their cities, temples, writing system, mathematics, and calendars.

Maya Terminal Classic - a time around the 9th century CE when many Maya cities in the southern lowlands were abandoned, often due to environmental or social challenges.

Mesoamerica - a cultural region stretching from central Mexico through parts of Central America. Ancient civilizations such as the Maya, Aztec, and Olmec lived there.

Radiocarbon dating - a method used by scientists to determine the age of ancient objects and remains by measuring the amount of carbon isotopes they contain.

Stalagmite - a rock formation that grows upward from the floor of a cave. It forms when water dripping from the ceiling leaves behind minerals such as calcite over a long time.

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Check your understanding

- 1 How many extreme droughts (lasting at least three years) did we find in the stalagmite record?
- 2 Why was it important to collect rainwater every month for a year?
- 3 How did we find out when the Classic Maya civilization stopped making monuments?
- 4 We think many Classic Maya people moved due to extreme droughts. We sometimes use the term "climate migration" to describe what happens when lots of people move because of changes to the climate. Other than droughts, what kinds of climate change might cause climate migration?
- 5 You can count the rings on a tree to learn its age, and you can study the layers in a stalagmite to learn about rain. Can you think of any other natural records?
- 6 The outside ring of a tree trunk is the youngest. The top layers of a stalagmite are the most recent. In a small group, imagine what a scientist might be able to figure out about your life based on the order of objects you have left or stored at home or at school. What intentional or accidental records do you create as you go about your own life?

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